

Library Management System Architecture

A comprehensive exploration of a C#-based library management system built with Visual Studio. This presentation breaks down the project structure, dependency relationships, and core functionality.



Project Structure Overview



Controller

LibraryManager orchestrates all operations and coordinates between services



Models

LibraryItem hierarchy with Book, EBook, Magazine, and User entities



Services

Authorization, borrowing, buying, and user management operations



Repository

Data persistence layer for items and users with CRUD operations

The system follows a layered architecture with clear separation of concerns, making it maintainable and scalable.

Core Controller: LibraryManager

Dependency Injection

The LibraryManager constructor injects four key services through dependency injection, ensuring loose coupling and testability.

01

Authorization Check

Verifies user permissions before executing any operation

03

Repository Call

Delegates data persistence to the repository layer

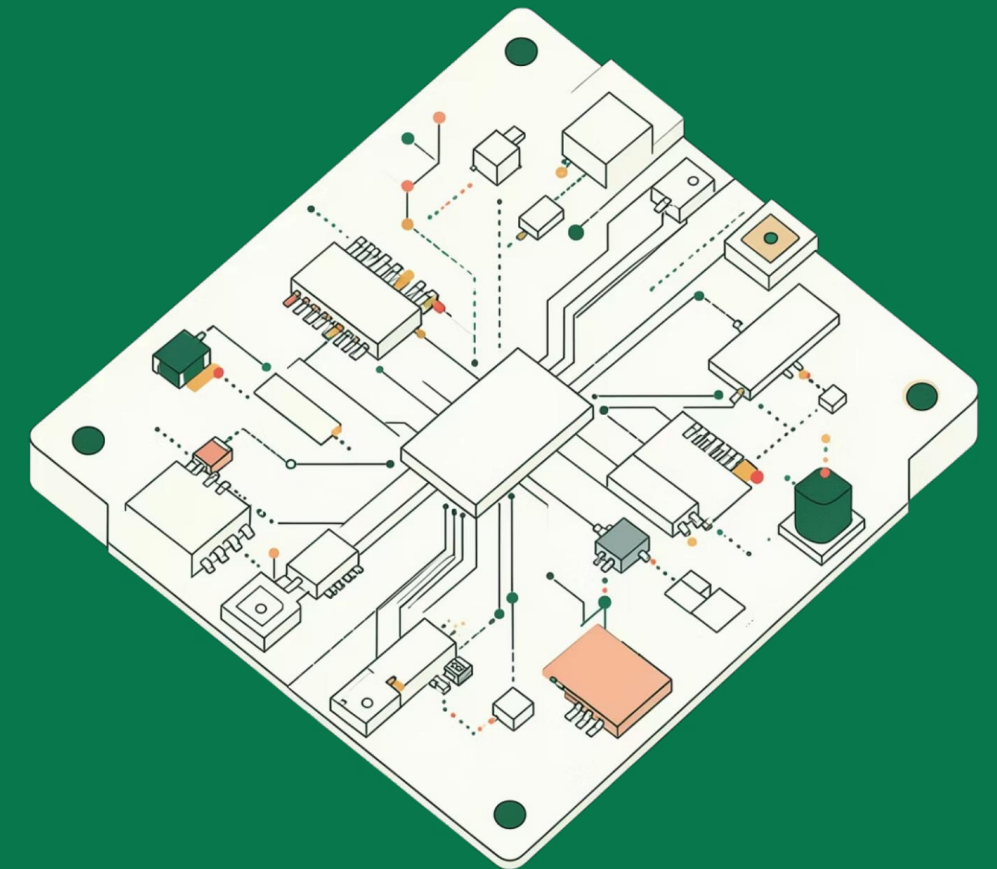
Central Orchestrator

Acts as the main coordinator for all library operations, handling permission checks and delegating to appropriate services.

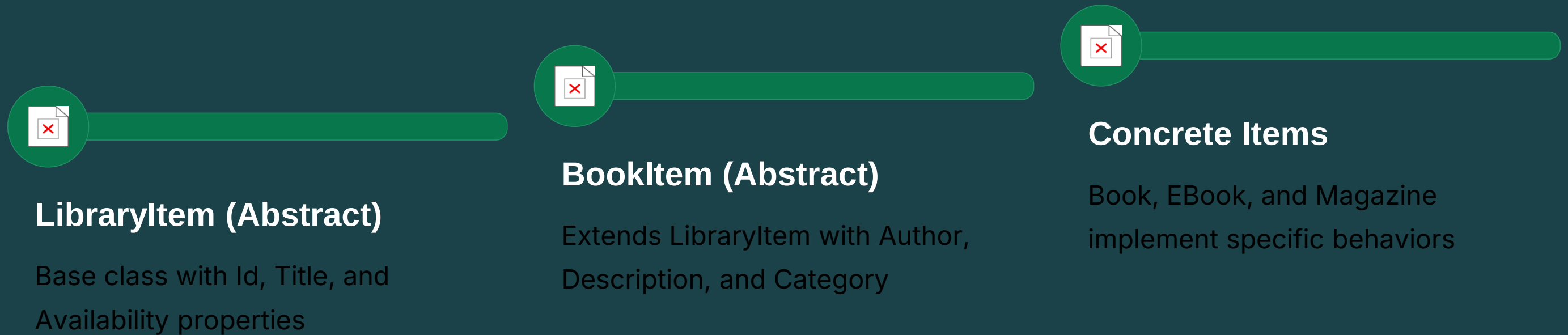
02

Business Logic

Processes the operation and validates data integrity



Model Hierarchy



The inheritance hierarchy promotes code reuse while allowing specialized functionality for each item type.

Service Layer Components



AuthService

Manages user permissions based on roles (User/Admin). Controls access to sensitive operations like adding or deleting items.



BuyingService

Processes purchases by checking availability and updating user's bought items list. Works with any IBuyable item.



BorrowingService

Manages borrowing with 14-day limits and late fees. Tracks due dates and calculates penalties for overdue books.



UserService

Handles user management operations with admin-only permissions. Manages user creation, updates, and deletions.

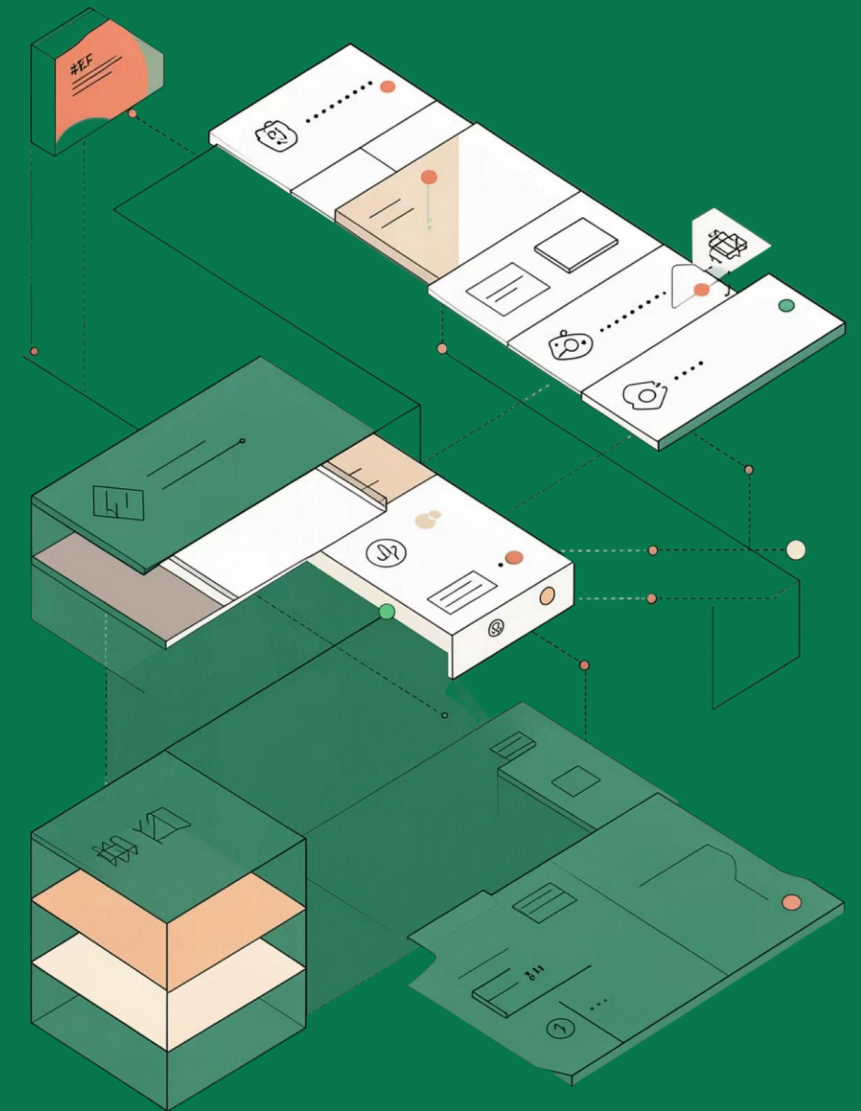
Repository Implementation

LibraryRepo

- In-memory storage using List
- CRUD operations with ID
- validation
- Prevents duplicate IDs
- Returns copies to maintain encapsulation

UserRepo

- Manages user collection with unique constraints
- Validates email format and ID uniqueness
- Tracks user roles and permissions
- Associates borrowed and purchased items



Key Features & Functionality

Role-Based Access Control

Admins can manage items while users can borrow and buy. Permission checks on every operation prevent unauthorized access.

Item Management

Add, update, delete, and search library items. Type-safe operations with comprehensive validation and error handling.

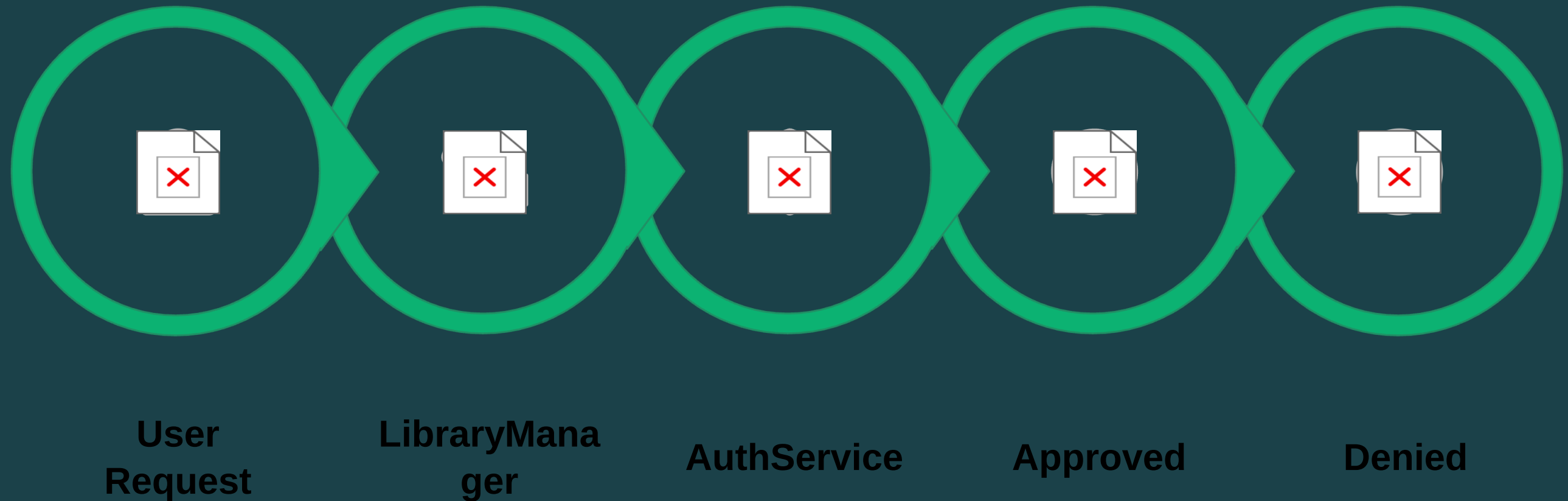
Borrowing System

Track due dates with automatic fine calculation. Enforce borrowing limits (3 books maximum per user).

Purchase System

Buy eBooks and magazines. Items become unavailable after purchase and are tracked in user's purchase history.

Authorization Flow



Every operation flows through authorization first. The AuthService validates user roles before proceeding to business logic, ensuring security throughout the system.

System Capabilities Summary

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Entity Types

LibraryItem, Book, EBook, Magazine, User

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Service Layers

Auth, Borrowing, Buying, User management

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Interface Contracts

IBorrowable, IBuyable, ILibraryRepo

2

User Roles

Admin with full access, User with limited permissions

This robust architecture provides a solid foundation for library management with proper separation of concerns, dependency injection, and comprehensive error handling.



Employee Demo:

https://drive.google.com/file/d/13limSZAjNo3KPNVBFk8cAmxvT7ej_iSK/view?usp=sharing

Admin Demo :

<drive.google.com/file/d/1BO6C4ncj0Plty5U9RJeolTigb2TzIZz0/view?usp=sharing>

User Demo :

<https://drive.google.com/file/d/1xs2p351Fm8h3oh0kCPFjMDwH86x-xUS9/view?usp=sharing>